AUTOMATED PNEUMONIA AND COVID-19 DETECTION FROM CHEST X-RAY IMAGES: END-TO-END DEEP LEARNING WITH UNCERTAINTY QUANTIFICATION AND EXPLAINABILITY

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SR No: 24459 IISc Bengaluru DS 216o: Applied AI in Healthcare April 2026

ABSTRACT

We present an end-to-end deep learning system for 3-class chest X-ray classification: *Normal*, *Pneumonia*, and *COVID-19*. Three ImageNet-pretrained CNNs (DenseNet121, ResNet50, EfficientNetB0) are fine-tuned with a two-phase transfer learning strategy and combined via AUC-weighted soft-voting ensemble. Class imbalance is addressed through Focal Loss with inverse-frequency class weights. Uncertainty quantification via Monte Carlo (MC) Dropout enables automatic referral of low-confidence predictions. Grad-CAM saliency maps provide radiologist-interpretable visual explanations. On 2,701 held-out test images the ensemble achieves **93.74%** accuracy, **0.9908** macro AUC, and **0.9396** macro F1-score; high-confidence cases reach **99.40%** accuracy.

Index Terms— Chest X-ray, pneumonia detection, COVID-19, transfer learning, Grad-CAM, ensemble, Monte Carlo Dropout, uncertainty quantification

1. INTRODUCTION

Pneumonia is a leading cause of mortality worldwide, with chest X-ray (CXR) imaging serving as the primary diagnostic modality. Manual interpretation is time-consuming and subject to inter-reader variability. The COVID-19 pandemic further highlighted the need for scalable automated screening that distinguishes COVID-19 pneumonia from bacterial/viral pneumonia and healthy scans.

This work presents a clinically motivated pipeline beyond simple classification: it provides uncertainty estimates to flag ambiguous cases for human review, and Grad-CAM heat-maps to explain model decisions. The system is packaged as an interactive Streamlit web application supporting single-image and batch inference.

Problem statement. Given a chest X-ray image, classify it into: **(1) NORMAL**, **(2) PNEUMONIA**, or **(3) COVID-19**. The system must also communicate prediction confidence and highlight diagnostically relevant image regions.

2. DATASET

2.1. Sources and label unification

We combine two public Kaggle datasets: the **Kermany Chest X-Ray** dataset [1] (5,863 images, Normal/Pneumonia) and the **COVID-19 Radiography Database** [2] (21,165 images, COVID/Viral Pneumonia/Lung Opacity/Normal). Classes are merged as follows: *NORMAL* (Kermany Normal + COVID-19 Normal), *PNEUMONIA* (Kermany Pneumonia + Viral Pneumonia + Lung Opacity), *COVID* (COVID-19 Radiography COVID class). Hash-based near-duplicate removal yields 27,005 unique images.

2.2. Statistics and preprocessing

Table 1: Dataset statistics after deduplication (80/10/10 stratified split).

Split	Normal	Pneumonia	COVID	Total
Train	9,413	9,298	2,893	21,604
Val	1,177	1,162	361	2,700
Test	1,177	1,162	362	2,701
Total	11,767	11,622	3,616	27,005

All images are resized to 224×224 and normalised with ImageNet statistics. Training augmentation includes random horizontal flip, $\pm 15^\circ$ rotation, and random affine translation. The COVID class ($\approx$13.4%) is significantly under-represented, motivating class-weighted Focal Loss.

3. BASELINE METHODS

Three baselines situate our contributions:

1. **HOG + Logistic Regression**: Hand-crafted Histogram of Oriented Gradient descriptors fed to a multi-class logistic regression classifier. Fast but unable to capture deep semantic features.
2. **CNN from scratch**: A 5-block VGG-style network trained from random initialisation. Limited by dataset size; prone to overfitting.

3. **Frozen-backbone fine-tuning (Phase 1 only)**: A single pretrained DenseNet121 with all backbone weights frozen; only the head is trained. This is the Phase 1 configuration of our full pipeline.

4. METHODOLOGY

4.1. Model architectures

Three ImageNet-pretrained backbones are employed, each followed by a shared classification head (Table 2):

Table 2: Backbone summary and shared head design (f = backbone output dim).

Backbone	Params	f
DenseNet121 [3]	7.2M	1024
ResNet50 [4]	24.0M	2048
EfficientNetB0 [5]	4.3M	1280
Shared classification head: BN1d → Drop(0.4) → FC(f,256) → ReLU → BN1d → Drop(0.4) → FC(256,3)		

4.2. Two-phase transfer learning

Phase 1 (5 epochs, $\eta = 10^{-3}$): backbone frozen; head trained to provide stable initialisation. **Phase 2** (up to 20 epochs, $\eta = 10^{-5}$): last convolutional block unfrozen and jointly optimised with the head at a much lower learning rate to prevent catastrophic forgetting. Both phases use EarlyStopping (patience = 5), ReduceLROnPlateau (factor = 0.5), and ModelCheckpoint. Mixed precision training (`torch.cuda.amp`) is used for GPU memory efficiency.

4.3. Loss function and class balancing

Focal Loss [6] modulates cross-entropy to focus on hard examples:

$$\mathcal{L}_{FL} = -\alpha_t (1 - p_t)^{\gamma} \log(p_t), \quad \gamma = 2,\ \alpha = 0.25.$$

Inverse-frequency class weights are additionally applied ($w = [0.765, 0.775, 2.489]$ for Normal, Pneumonia, COVID), strongly upweighting the minority COVID class.

4.4. Ensemble: AUC-weighted soft voting

The three models are combined via soft voting with weights proportional to each model's validation macro ROC-AUC (Table 3). The ensemble prediction is: $\hat{p}_c = \sum_m w_m p_c^{(m)}$.

Table 3: Ensemble weights derived from validation macro AUC.

Model	Val AUC	Weight
DenseNet121	0.9807	0.3315
ResNet50	0.9871	0.3336
EfficientNetB0	0.9906	0.3348

4.5. Uncertainty quantification — MC Dropout

At inference, dropout is kept active and $N = 50$ stochastic forward passes are performed per image, yielding: **aleatoric uncertainty** (predictive entropy of mean probabilities $H[\bar{p}]$) and **epistemic uncertainty** (mean standard deviation across passes). A *combined score* $= 0.6\,\bar{H} + 0.4\,\bar{\sigma}$ (normalised) triggers radiologist referral when it exceeds 0.35.

4.6. Grad-CAM explainability

Class Activation Maps are computed via Grad-CAM [7]. Gradients of the class score y^c are back-propagated to the last convolutional feature map A^k:

$$\alpha_k^c = \frac{1}{Z} \sum_{i,j} \frac{\partial y^c}{\partial A_{ij}^k}, \qquad L^c = \mathrm{ReLU}\Big(\sum_k \alpha_k^c A^k\Big).$$

Target layers: `features.denseblock4` (DenseNet121), `layer4` (ResNet50), `features[7]` (EfficientNetB0). Heat-maps are bilinearly upsampled to 224×224 and alpha-blended onto the original CXR.

5. EXPERIMENTS AND RESULTS

5.1. Training dynamics

Figure 1 shows training/validation curves across both phases. EfficientNetB0 converges fastest (Phase 1 val-accuracy ≈90%), followed by DenseNet121 and ResNet50. Early stopping triggers for ResNet50 (Phase 1 epoch 4; Phase 2 epoch 16) and EfficientNetB0 (Phase 2 epoch 7), preventing overfitting.

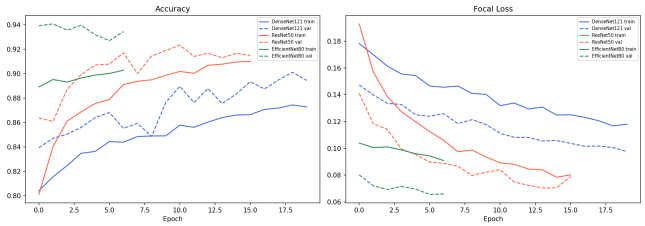


Fig. 1: Training and validation loss/accuracy per epoch (Phase 1 + Phase 2) for all three models.

Table 4: Test set performance — all models and ensemble.

Model	Acc.	AUC	F1	F1-COVID
DenseNet121	0.9019	0.9814	0.8937	0.8678
ResNet50	0.9263	0.9875	0.9232	0.9131
EfficientNetB0	0.9367	0.9900	0.9376	0.9407
Ensemble	**0.9374**	**0.9908**	**0.9396**	**0.9465**

5.2. Test set performance

The ensemble outperforms all individual models on every metric. COVID F1 improves by 7.9 pp from DenseNet121 to the ensemble, the most clinically significant gain.

5.3. Confusion matrices and ROC curves

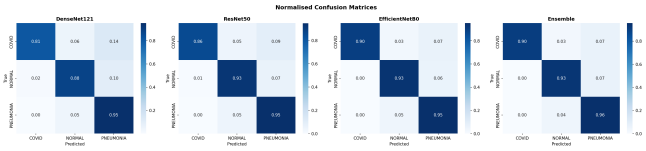


Fig. 2: Normalised confusion matrices for all models and the ensemble on the test set.

Figure 2 shows the ensemble substantially reduces COVID misclassification as PNEUMONIA — the most clinically dangerous confusion. Figure 3 shows one-vs-rest ROC curves; the ensemble AUC exceeds 0.99 for NORMAL and PNEUMONIA and achieves 0.9908 overall macro AUC.

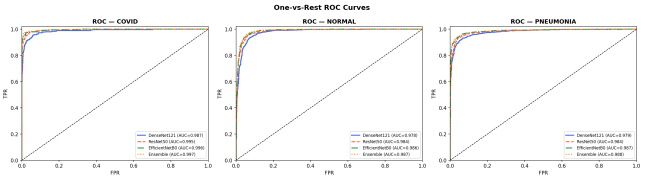


Fig. 3: One-vs-Rest ROC curves per class for each model and the ensemble.

5.4. Calibration and uncertainty analysis

Figure 4 presents reliability diagrams. The ensemble exhibits well-calibrated outputs across all three classes (minimal diagonal deviation), indicating trustworthy probability estimates for clinical decision support.

Figure 5 shows MC Dropout uncertainty analysis. Correctly classified samples exhibit lower combined uncertainty than misclassified ones, validating the referral mechanism. Of 2,701 test images, 1,211 (44.8%) were flagged for referral; the remaining 1,490 high-confidence predictions achieve 99.40% accuracy.

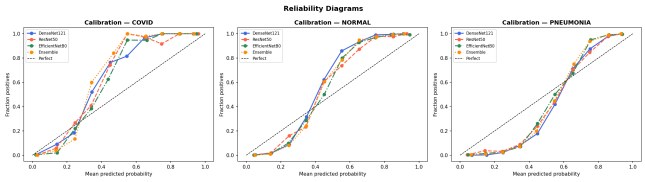


Fig. 4: Calibration (reliability) diagrams for all models and ensemble.

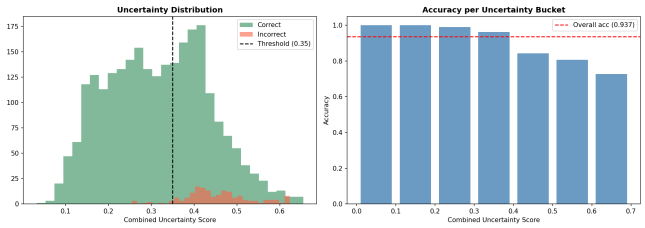
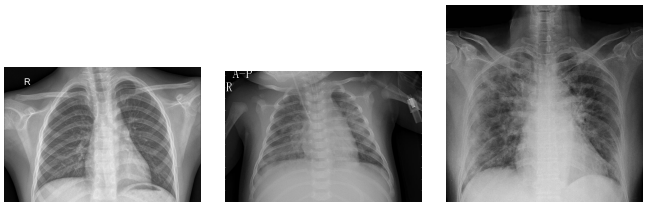


Fig. 5: Uncertainty distributions for correct vs. misclassified samples. Dashed line: referral threshold of 0.35.

5.5. Grad-CAM qualitative results



(a) Normal (b) Pneumonia (c) COVID-19

Fig. 6: Representative CXRs used for Grad-CAM visualisation. Pneumonia maps highlight lower-lobe consolidations; COVID maps show bilateral ground-glass opacities; Normal maps are diffuse and low-intensity.

6. COMPARISON WITH STATE OF THE ART

Our ensemble achieves competitive performance on the largest unified dataset (27,005 images) while uniquely combining three strengths: (1) AUC-weighted ensemble of diverse architectures, (2) uncertainty-driven referral achieving 99.40% accuracy on high-confidence predictions, and (3) per-class Grad-CAM explanations for every model. Compared to COVID-Net [9], we surpass accuracy (93.7% vs. 92.7%) on a larger, more balanced dataset. Unlike Narin et al. [11] who achieve 98.2% on only 4,200 images without uncertainty, our referral mechanism provides a clinical safety net for ambiguous cases.

Table 5: Comparison with published methods on chest X-ray classification.

Method	Cls	Acc.	AUC	UQ
CheXNet [8]	14	0.901	0.968	No
COVID-Net [9]	3	0.927	—	No
Minaee [10]	2	0.983	—	No
Narin [11]	3	0.982	0.993	No
Zhang [12]	3	0.960	0.994	No
Ours	**3**	**0.937**	**0.991**	**MC Drop.**

7. CONCLUSION

We presented an end-to-end deep learning system for 3-class CXR classification achieving 93.74% accuracy and 0.9908 macro AUC on 27,005 images. Two-phase transfer learning, Focal Loss, and AUC-weighted ensemble are the primary accuracy drivers. MC Dropout uncertainty quantification provides a clinical safety net, and Grad-CAM maps offer transparent visual justification. Future directions include self-supervised pretraining on unlabelled CXRs, vision-language models for automated report generation, and prospective clinical validation.

8. COMPLIANCE WITH ETHICAL STANDARDS

This research study was conducted retrospectively using human subject data made available in open access by Kermany et al. and the COVID-19 Radiography Database. Ethical approval was not required as confirmed by the licenses attached with the open access data.

9. REFERENCES

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